Lab 11: Protein Structure Prediction with AlphaFold

Abigail Perez

Overview

In this lab, we used AlphaFold2 via ColabFold to predict the structure of HIV Protease (HIV-Pr), first as a monomer and then as a homodimer. We analyzed the resulting models using Bio3D in R, examining structural consistency, confidence scores (pLDDT), predicted aligned error (PAE), and residue conservation.

Section 8: Custom Analysis of AlphaFold Dimer Models

All analysis below uses the HIV-Pr homodimer ColabFold results (`HIV_Pr_dimer_23119/`).

Loading PDB Files

```
library(bio3d)

# 1. Load files
results_dir <- "HIV_Pr_dimer_23119/"
pdb_files <- list.files(path=results_dir, pattern="*.pdb", full.names=TRUE)
basename(pdb_files)
```

```
[1] "HIV_Pr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIV_Pr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIV_Pr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIV_Pr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIV_Pr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

Aligning and Fitting Models

```r
# 2. Align and fit
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed
HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed
HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed
HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed
HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed
.....

Extracting sequences

pdb/seq: 1   name: HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_001_alphafold2_multi
pdb/seq: 2   name: HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_002_alphafold2_multi
pdb/seq: 3   name: HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_003_alphafold2_multi
pdb/seq: 4   name: HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_004_alphafold2_multi
pdb/seq: 5   name: HIV_Pr_dimer_23119//HIV_Pr_dimer_23119_unrelaxed_rank_005_alphafold2_multi
```

```r
pdbs
```

```
                                1        .         .         .         .         50
[Truncated_Name:1]HIV_Pr_dim   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIV_Pr_dim   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIV_Pr_dim   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIV_Pr_dim   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIV_Pr_dim   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                               **************************************************
                               1        .         .         .         .         50

                               51       .         .         .         .         100
[Truncated_Name:1]HIV_Pr_dim   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]HIV_Pr_dim   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]HIV_Pr_dim   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]HIV_Pr_dim   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]HIV_Pr_dim   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                               **************************************************
                               51       .         .         .         .         100
```

```
                             101        .         .         .         .         150
[Truncated_Name:1]HIV_Pr_dim   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]HIV_Pr_dim   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]HIV_Pr_dim   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]HIV_Pr_dim   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]HIV_Pr_dim   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                               **************************************************
                             101        .         .         .         .         150

                             151        .         .         .         .       198
[Truncated_Name:1]HIV_Pr_dim   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIV_Pr_dim   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIV_Pr_dim   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]HIV_Pr_dim   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIV_Pr_dim   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               **************************************************
                             151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

The alignment shows 5 sequence rows with 198 position columns (99 residues × 2 chains), with no gaps since all models have identical sequences.

RMSD Analysis

RMSD (Root Mean Square Deviation) measures structural distance between coordinate sets.

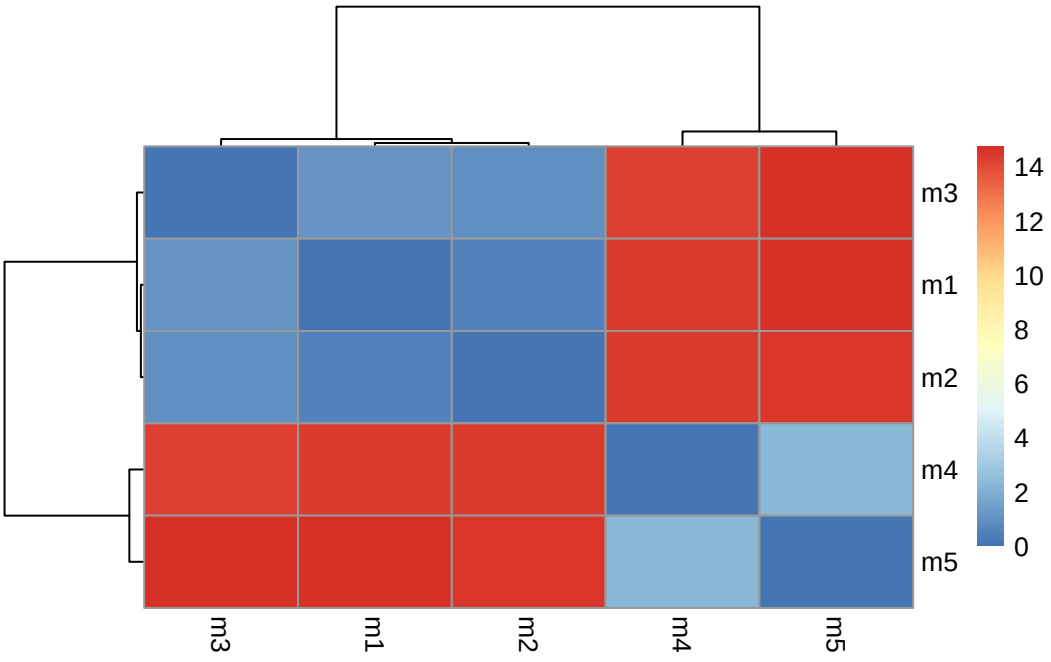
```r
# 3. RMSD matrix
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.754
```

```r
# 4. RMSD heatmap
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



The heatmap shows that models 1 and 2 are structurally similar to each other, while models 4 and 5 are quite different — consistent with their lower average pLDDT scores (76.57 and 72.98 respectively).

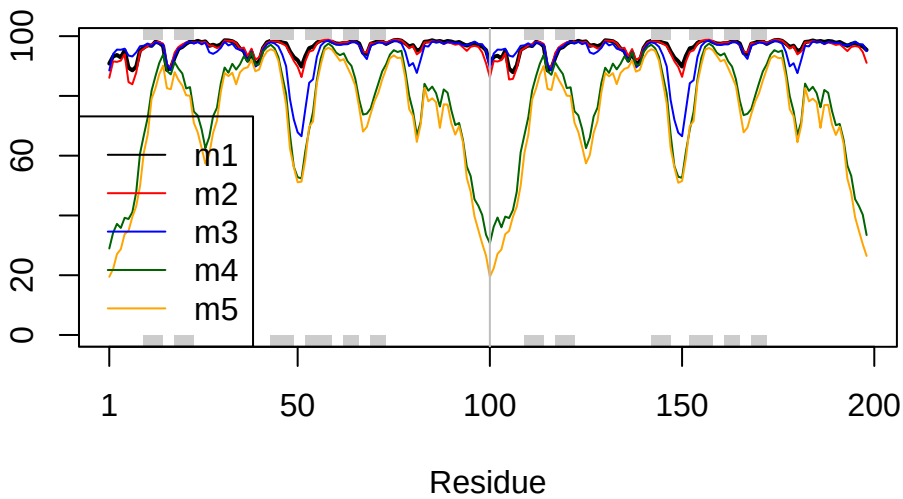
pLDDT Scores Across Models

pLDDT (predicted local distance difference test) is a per-residue confidence score ranging from 0–100. Values above 70 are considered reliable; below 50 are unreliable.

```r
# 5. pLDDT plot
pdb <- read.pdb("1hsg")
```

```
Note: Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
legend("bottomleft", legend=paste0("m",1:5),
       col=c("black","red","blue","darkgreen","orange"), lty=1)
```



The vertical gray line at position 100 marks the boundary between chain A and chain B. Models 1–3 maintain high pLDDT throughout, while models 4 and 5 show notably lower confidence, particularly in chain B.

Core Superposition

To improve structural comparison, we identify the most consistent rigid core across all models.

```
# 6. Core superposition
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 9885.419
core size 196 of 198  vol = 6898.241
core size 195 of 198  vol = 1338.035
core size 194 of 198  vol = 1040.677
core size 193 of 198  vol = 951.865
core size 192 of 198  vol = 899.087
core size 191 of 198  vol = 834.733
core size 190 of 198  vol = 771.342
core size 189 of 198  vol = 733.069
core size 188 of 198  vol = 697.285
core size 187 of 198  vol = 659.748
core size 186 of 198  vol = 625.28
core size 185 of 198  vol = 589.548
core size 184 of 198  vol = 568.261
core size 183 of 198  vol = 545.022
core size 182 of 198  vol = 512.897
core size 181 of 198  vol = 490.731
core size 180 of 198  vol = 470.274
core size 179 of 198  vol = 450.738
core size 178 of 198  vol = 434.743
core size 177 of 198  vol = 420.345
core size 176 of 198  vol = 406.666
core size 175 of 198  vol = 393.341
core size 174 of 198  vol = 382.402
core size 173 of 198  vol = 372.866
core size 172 of 198  vol = 357.001
core size 171 of 198  vol = 346.576
core size 170 of 198  vol = 337.454
core size 169 of 198  vol = 326.668
core size 168 of 198  vol = 314.959
core size 167 of 198  vol = 304.136
core size 166 of 198  vol = 294.561
core size 165 of 198  vol = 285.658
core size 164 of 198  vol = 278.893
core size 163 of 198  vol = 266.773
core size 162 of 198  vol = 259.003
core size 161 of 198  vol = 247.731
core size 160 of 198  vol = 239.849
core size 159 of 198  vol = 234.973
core size 158 of 198  vol = 230.071
core size 157 of 198  vol = 221.995
core size 156 of 198  vol = 215.629
core size 155 of 198  vol = 206.8
```

```
core size 154 of 198  vol = 196.992
core size 153 of 198  vol = 188.547
core size 152 of 198  vol = 182.27
core size 151 of 198  vol = 176.961
core size 150 of 198  vol = 170.72
core size 149 of 198  vol = 166.128
core size 148 of 198  vol = 159.805
core size 147 of 198  vol = 153.775
core size 146 of 198  vol = 149.101
core size 145 of 198  vol = 143.664
core size 144 of 198  vol = 137.145
core size 143 of 198  vol = 132.523
core size 142 of 198  vol = 127.237
core size 141 of 198  vol = 121.579
core size 140 of 198  vol = 116.78
core size 139 of 198  vol = 112.575
core size 138 of 198  vol = 108.175
core size 137 of 198  vol = 105.137
core size 136 of 198  vol = 101.254
core size 135 of 198  vol = 97.379
core size 134 of 198  vol = 92.978
core size 133 of 198  vol = 88.188
core size 132 of 198  vol = 84.032
core size 131 of 198  vol = 81.902
core size 130 of 198  vol = 78.023
core size 129 of 198  vol = 75.276
core size 128 of 198  vol = 73.057
core size 127 of 198  vol = 70.699
core size 126 of 198  vol = 68.976
core size 125 of 198  vol = 66.707
core size 124 of 198  vol = 64.376
core size 123 of 198  vol = 61.145
core size 122 of 198  vol = 59.029
core size 121 of 198  vol = 56.625
core size 120 of 198  vol = 54.369
core size 119 of 198  vol = 51.826
core size 118 of 198  vol = 49.651
core size 117 of 198  vol = 48.19
core size 116 of 198  vol = 46.644
core size 115 of 198  vol = 44.748
core size 114 of 198  vol = 43.288
core size 113 of 198  vol = 41.089
core size 112 of 198  vol = 39.143
```

```
 core size 111 of 198  vol = 36.468
 core size 110 of 198  vol = 34.114
 core size 109 of 198  vol = 31.467
 core size 108 of 198  vol = 29.445
 core size 107 of 198  vol = 27.323
 core size 106 of 198  vol = 25.82
 core size 105 of 198  vol = 24.149
 core size 104 of 198  vol = 22.647
 core size 103 of 198  vol = 21.068
 core size 102 of 198  vol = 19.953
 core size 101 of 198  vol = 18.3
 core size 100 of 198  vol = 15.723
 core size 99 of 198  vol = 14.841
 core size 98 of 198  vol = 11.646
 core size 97 of 198  vol = 9.435
 core size 96 of 198  vol = 7.354
 core size 95 of 198  vol = 6.181
 core size 94 of 198  vol = 5.667
 core size 93 of 198  vol = 4.706
 core size 92 of 198  vol = 3.664
 core size 91 of 198  vol = 2.77
 core size 90 of 198  vol = 2.151
 core size 89 of 198  vol = 1.715
 core size 88 of 198  vol = 1.15
 core size 87 of 198  vol = 0.874
 core size 86 of 198  vol = 0.685
 core size 85 of 198  vol = 0.528
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

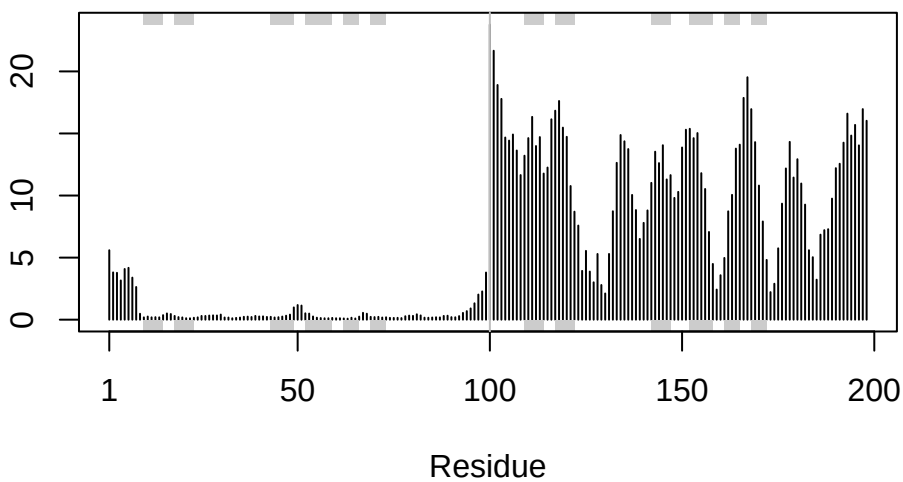
```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

RMSF Analysis

RMSF (Root Mean Square Fluctuation) measures positional variability across models at each residue.

```
# 7. RMSF
rf <- rmsf(xyz)
plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Chain A (positions 1–99) is largely consistent across models. Chain B (positions 100–198) shows much greater variability, indicating that the second chain is less reliably predicted — a known limitation of AlphaFold monomer-based dimer modeling.

Predicted Aligned Error (PAE)

PAE measures confidence in the relative position of pairs of residues. Low PAE = high confidence.

```
# 8. PAE from JSON files
library(jsonlite)
pae_files <- list.files(path=results_dir, pattern=".*model.*\\.json", full.names=TRUE)
```

```r
pae1 <- read_json(pae_files[1], simplifyVector=TRUE)
pae5 <- read_json(pae_files[5], simplifyVector=TRUE)
```

```r
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

```r
# Maximum PAE values (lower = better)
pae1$max_pae
```

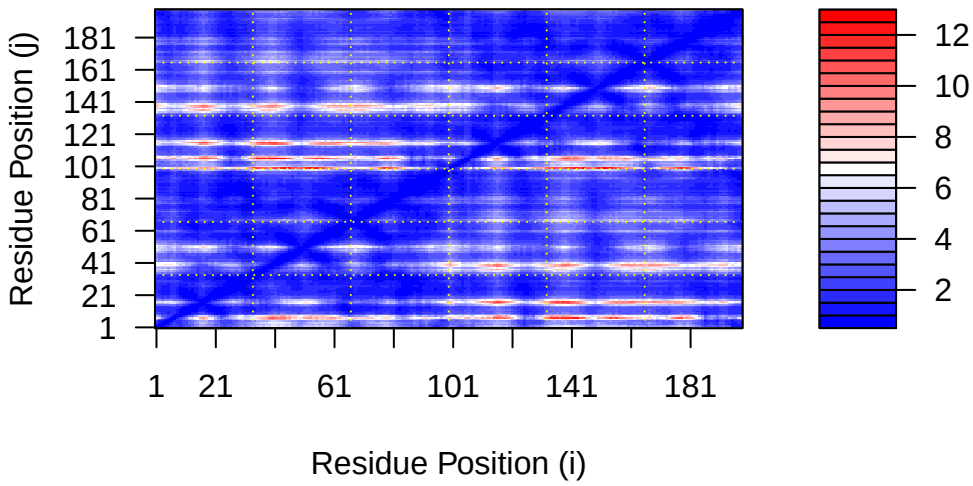
```
[1] 12.84375
```

```r
pae5$max_pae
```

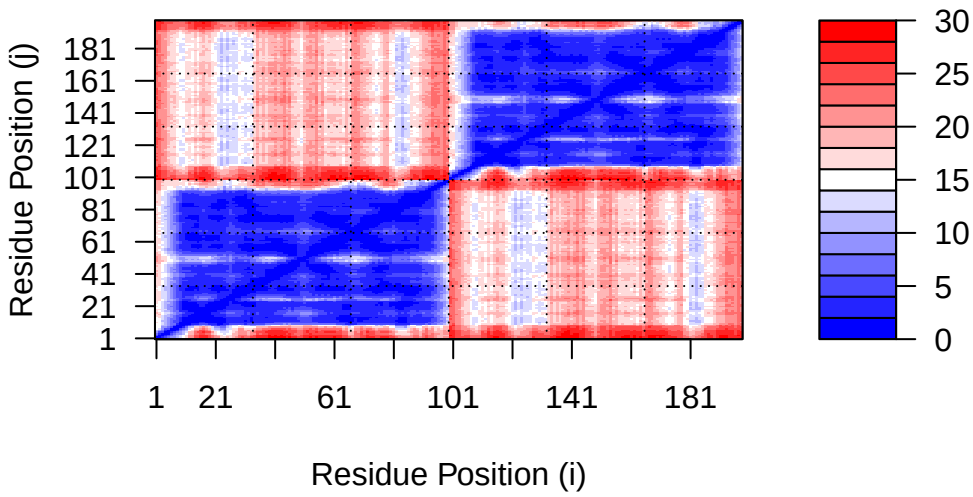
```
[1] 29.59375
```

Model 1 has a much lower max PAE than model 5, confirming it is the higher quality prediction.

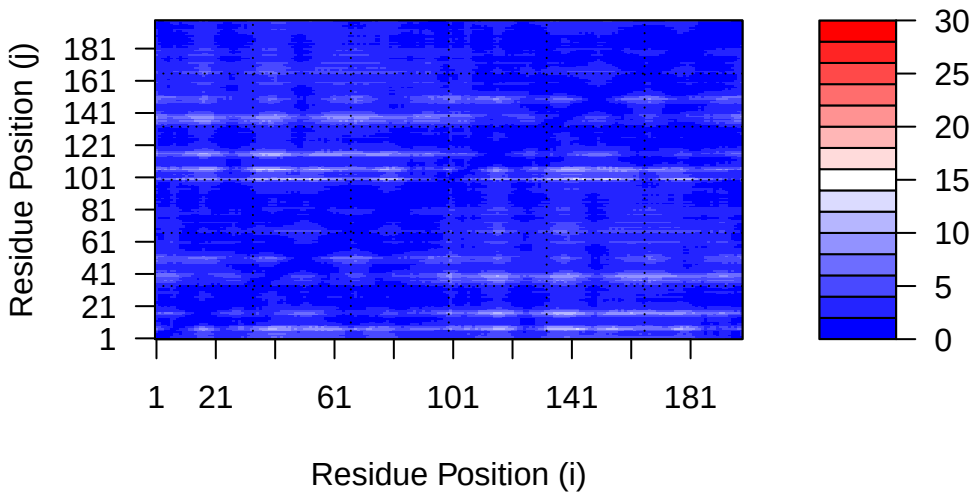
```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Model 1's PAE plot shows clear off-diagonal dark blocks, indicating confident inter-domain packing between the two chains. Model 5 shows a more uniform (lighter) matrix, indicating poor confidence in the relative positioning of the two chains.

Residue Conservation

We read the MSA generated by ColabFold and score conservation at each position.

```
# 9. Residue conservation
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "HIV_Pr_dimer_23119//HIV_Pr_dimer_23119.a3m"
```

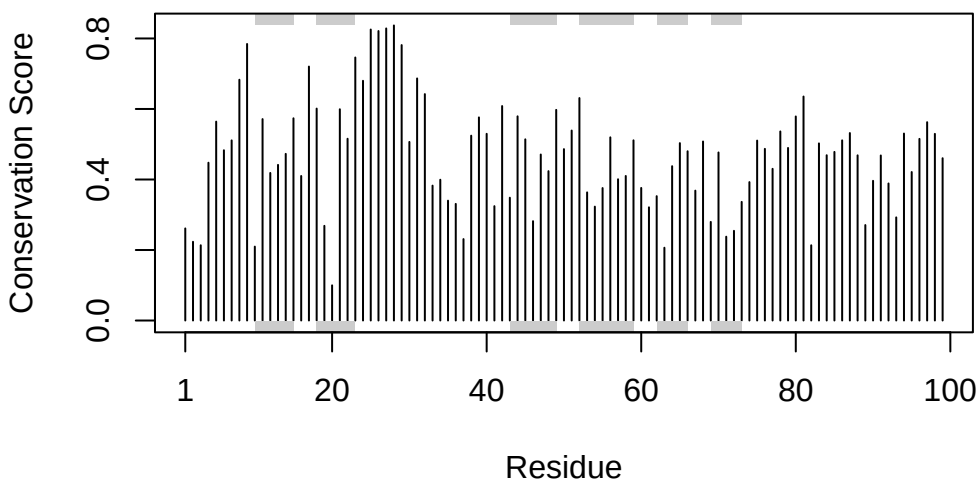
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```r
dim(aln$ali)
```

```
[1] 5397  132
```

```r
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

The consensus sequence highlights the conserved active site residues **D25, T26, G27, A28** (the DTGA motif), which are critical for HIV protease catalytic activity.

Writing Conservation to PDB for Mol* Visualization

```
# 10. Write conservation PDB for Mol*
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

Mol* Visualizations

The following figures were generated in Mol* using the ColabFold dimer PDB output files.

Figure 18: Superposed Dimer Models Colored by pLDDT

Figure 1: Superposed HIV-Pr dimer models colored by pLDDT (Uncertainty/Disorder). Red = very high confidence (pLDDT > 90). The variable flap regions visible at the top show slightly lower confidence.

Figure 19: Core-Fitted Structures Colored by pLDDT

Figure 2: Core-fitted dimer models colored by pLDDT. Models were superposed using the rigid core identified by `core.find()` in Bio3D. The two-chain dimer structure is clearly visible.

Figure 20: Top Model Colored by Sequence Conservation

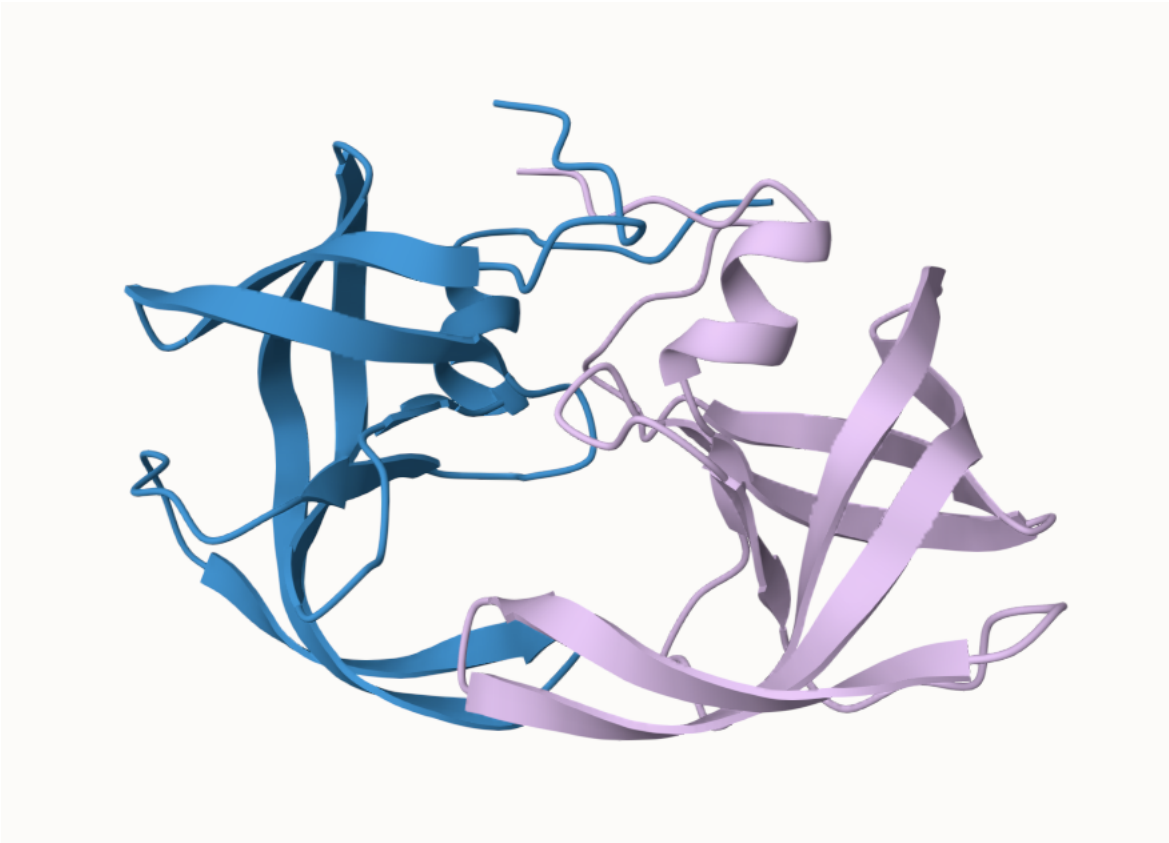


Figure 3: Top-ranked dimer model (m1_conserv.pdb) colored by residue conservation (Occupancy column). Teal regions are more conserved; orange regions are less conserved. The conserved DTGA active site motif is visible at the interface between the two chains.

Session Info

```r
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Sonoma 14.8.2

Matrix products: default
```

```
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] jsonlite_2.0.0  pheatmap_1.0.13 bio3d_2.4-5

loaded via a namespace (and not attached):
 [1] crayon_1.5.3        cli_3.6.6           knitr_1.51
 [4] rlang_1.2.0         xfun_0.57           otel_0.2.0
 [7] generics_0.1.4      glue_1.8.1          S4Vectors_0.48.1
[10] Biostrings_2.78.0   htmltools_0.5.9     stats4_4.5.3
[13] scales_1.4.0        rmarkdown_2.31      grid_4.5.3
[16] Seqinfo_1.0.0       evaluate_1.0.5      fastmap_1.2.0
[19] lifecycle_1.0.5     yaml_2.3.12         IRanges_2.44.0
[22] compiler_4.5.3      RColorBrewer_1.1-3  Rcpp_1.1.1-1.1
[25] XVector_0.50.0      rstudioapi_0.18.0   farver_2.1.2
[28] digest_0.6.39       R6_2.6.1            parallel_4.5.3
[31] gtable_0.3.6        tools_4.5.3         msa_1.42.0
[34] BiocGenerics_0.56.0
```

References

Jumper, J. et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, 596, 583–589.

Mirdita, M. et al. (2022). ColabFold: Making protein folding accessible to all. *Nature Methods*, 19, 679–682.

Baek, M. et al. (2021). Accurate prediction of protein structures and interactions using a three-track neural network. *Science*, 373, 871–876.